Reconstruction Without Pairwise Distances:
A Large-Scale Comparison of Autoencoders, DiRe, and UMAP

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Abstract

Graph-based dimension-reduction methods explicitly construct and optimize relationships between observations. A reconstruction autoencoder instead learns from individual input vectors using only mean-squared error. We compare these approaches at target widths $d \in \{2, 8\}$ on 723,457 arXiv paper embeddings and one million Wikipedia paragraph embeddings. All reducers receive the same row-normalized data and are evaluated over three seeds using sample-conditional neighborhood fidelity, sampled global-distance fidelity, local decodability, label- or group-aware diagnostics, sampled persistence-derived diagnostics, and synchronized fit time.

No method dominates the completed cells. At arXiv width two, the autoencoder leads the rank-aware local scores, UMAP gives the best category prediction, and DiRe gives the lowest sampled Betti-curve discrepancies. At width eight, the autoencoder leads every reported geometric score while UMAP still gives the best category prediction. On the Wikipedia corpus, width two divides the geometric columns among all three methods, while at width eight the autoencoder leads the local, distance-order, and decoding scores but not stress; DiRe gives the lowest global-distance stress at both widths and UMAP most strongly groups paragraphs from the same article. A separate controlled check on manifolds of known dimension finds that rounded TwoNN and autoencoder-elbow estimates agree exactly in four of nine seeded trials and are within one dimension in all nine, with interpretable differences between local and global reconstructive dimension.

The sampled topology comparison is similarly conditional on corpus and width. DiRe leads both Betti-curve discrepancies for arXiv at width two; at width eight the autoencoder leads β_0 and DiRe leads β_1. On Wikipedia, DiRe leads β_0 and the autoencoder leads β_1 at width two, while the autoencoder leads both at width eight. A paired BF16-autocast sensitivity check preserves reconstruction and shared-metric quality under prespecified practical margins. It nearly halves Wikipedia training time, but slightly slows the smaller arXiv network, showing that mixed-precision speedup depends on whether dense computation amortizes its overhead.

1 Why compare pointwise reconstruction with graph methods?

For a fixed architecture and epoch budget, an autoencoder's work is linear in the number of observations [1]:

$$O\left(EN \sum_{\ell} w_\ell w_{\ell+1}\right),$$

where E is the number of epochs and w_ℓ are layer widths. It never constructs a pairwise distance matrix or a neighborhood graph. DiRe and UMAP [3, 2], by contrast, begin from neighbor relationships. Exact neighbor construction would have quadratic scaling, so their scalable implementations use approximate-nearest-neighbor indices and then optimize a sparse graph. That avoids N^2 storage and work, but introduces indexing, sampling, and irregular memory access. Dense autoencoder matrix multiplications are particularly favorable on a large GPU.

This computational distinction does not imply that reconstruction preserves the geometry needed by a particular application. The central empirical question is therefore not whether one method has a universally better score, but which structure each objective retains.

Corpus	Rows	Input D	Target d	Label field	Label interpretation
arXiv	723,457	384	$2, 8$	primary category	subject category
Wikipedia	1,000,000	1,024	$2, 8$	article URL	same-article group

Table 1: The two large embedding corpora. Wikipedia URLs are group identifiers, not a topic taxonomy analogous to arXiv categories.

2 Data, reducers, and protocol

The arXiv matrix contains mean-pooled `BAAI/bge-small-en-v1.5` paper embeddings [9]. The Wikipedia matrix is the first million rows of the English configuration of `CohereLabs/wikipedia-2023-11-embed-multilingual-v3` [10].

Every input row is converted to float32 and normalized to unit Euclidean norm. Each method receives the complete normalized matrix at both $d = 2$ and $d = 8$, with reducer/training seeds 0, 1, and 2. Evaluation samples and sampled pairs are held fixed across methods, target widths, and seeds. DiRe and UMAP fitting, and all neighborhood-based evaluations, use $k = 16$; autoencoder fitting has no neighborhood parameter. The comparisons are transductive: every reported layout is fitted on the full corpus being evaluated.

Autoencoder. For encoder f_θ and decoder g_ϕ, training minimizes only elementwise reconstruction MSE,

$$\mathcal{L}(\theta, \phi) = \frac{1}{ND} \sum_{i=1}^{N} \|x_i - g_\phi(f_\theta(x_i))\|_2^2 . \tag{1}$$

There is no label, pairwise-distance, neighborhood, contrastive, or topology term. Hidden layers use SiLU activations; bottleneck and output layers are linear. The architectures are

$$384 \to 256 \to 128 \to 32 \to d \to 32 \to 128 \to 256 \to 384$$

for arXiv and

$$1024 \to 512 \to 256 \to 64 \to d \to 64 \to 256 \to 512 \to 1024$$

for Wikipedia. As d changes from two to eight, their parameter counts range from 271,682 to 272,072 and from 1,346,434 to 1,347,208, respectively. AdamW uses learning rate 10^{-3} and weight decay 10^{-5}, with batches of 32,768 rows. The primary comparison uses float32 arithmetic; a separate BF16 sensitivity arm is reported below.

Five percent of the rows select an epoch count under a 200-epoch cap and patience 15. A freshly initialized model is then refit on all rows for the selected number of epochs. The autoencoder is neither pretrained nor recycled between corpora, target widths, or seeds.

DiRe and UMAP. DiRe 0.3.1 uses its cuVS implementation, with PCA initialization, 128 layout iterations, and an automatically selected approximate-neighbor index. cuML UMAP 26.06 uses Euclidean distance, spectral initialization, `min_dist`= 0.1, and `spread`= 1. Native UMAP inverse transformation is disabled because its CPU Delaunay stage is not a scale-safe operation for these corpora.

Hardware and timing. Reducer runs use a single NVIDIA GH200, RAPIDS/cuML 26.06, PyTorch 2.11, and three independent reducer/training seeds. Persistence calculations use Ripser 0.6.14 and FastDTW 0.3.4 in the RAPIDS 26.04 environment because those packages were unavailable in the 26.06 environment. CUDA synchronization brackets every reported fit. DiRe and UMAP time their synchronized `fit_transform`; autoencoder time includes GPU staging, validation-based training, fresh full-data refitting, full-corpus encoding, and native reconstruction statistics. Source loading, row normalization, library import, downstream comparison metrics, and the final layout-file flush are outside "fit." One-time data preparation took 2.58 seconds for arXiv and 10.05–14.59 seconds across the two Wikipedia runs; it may be reused across target widths.

3 Evaluation metrics

Let $X = \{x_i\} \subset \mathbb{R}^D$ be the normalized input and $Y = \{y_i\} \subset \mathbb{R}^d$ a layout. Neighborhood metrics use Euclidean distance and $k = 16$. Their $O(n_s^2)$ rank calculations are performed within a declared fixed sample, not the complete corpus: a label-stratified 5,000-paper sample for arXiv and a uniform 3,000-paragraph sample for Wikipedia. Wikipedia has 153,264 URL groups, more than the number of sample slots, so label stratification is not possible. Identical sample indices are used for all methods, target widths, and reducer seeds.

3.1 Neighborhood fidelity

For a sampled point i, let $N_X^k(i)$ and $N_Y^k(i)$ be its k nearest neighbors in input and layout space, and let $r_X(i,j)$ and $r_Y(i,j)$ be the corresponding full within-sample ranks. Define the introduced and lost neighbor sets

$$U_k(i) = N_Y^k(i) \setminus N_X^k(i), \qquad V_k(i) = N_X^k(i) \setminus N_Y^k(i).$$

Trustworthiness [6] penalizes false neighbors introduced by the layout:

$$T(k) = 1 - \frac{2}{n_s k(2n_s - 3k - 1)} \sum_{i=1}^{n_s} \sum_{j \in U_k(i)} \big(r_X(i,j) - k\big). \tag{2}$$

Continuity is reversed trustworthiness and penalizes input neighbors lost by the layout:

$$C(k) = 1 - \frac{2}{n_s k(2n_s - 3k - 1)} \sum_{i=1}^{n_s} \sum_{j \in V_k(i)} \big(r_Y(i,j) - k\big). \tag{3}$$

Both approach one for strong local fidelity. Exact neighbor overlap is

$$O(k) = \frac{1}{n_s k} \sum_{i=1}^{n_s} \left| N_X^k(i) \cap N_Y^k(i) \right|. \tag{4}$$

Unlike trustworthiness and continuity, overlap gives no partial credit when a missed or introduced neighbor has nearly the correct rank.

3.2 Sampled global distances

For 200,000 fixed, independently sampled non-self pairs P, with pair repetitions allowed, write

$$\delta_{ij} = \|x_i - x_j\|_2, \qquad d_{ij} = \|y_i - y_j\|_2.$$

Pair Spearman correlation ρ_S is the ordinary correlation of the midranks of $\{\delta_{ij}\}$ and $\{d_{ij}\}$. The scale-aligned normalized stress is

$$S = \left[\frac{\sum_{(i,j)\in P} (\delta_{ij} - a d_{ij})^2}{\sum_{(i,j)\in P} \delta_{ij}^2} \right]^{1/2}, \qquad a = \frac{\sum_{(i,j)\in P} \delta_{ij} d_{ij}}{\sum_{(i,j)\in P} d_{ij}^2}. \tag{5}$$

Higher ρ_S and lower S are preferable. Equation (5) is a through-origin least-squares scale alignment; it is not Kruskal Stress-1 or a full Procrustes fit.

3.3 Shared local-decoding probe

A separate sample—10,000 rows for arXiv and 5,000 for Wikipedia—is split 80/20. The training layout is whitened, retaining covariance eigenvectors with $\lambda_j > \max(10^{-8}\lambda_{\max}, \epsilon)$. A distance-weighted 16-neighbor regressor predicts all coordinates of x from the whitened layout. With training and test folds A and B,

$$\text{DecodeNMSE} = \frac{\sum_{i\in B} \|x_i - \widehat{x}(y_i)\|_2^2}{\sum_{i\in B} \|x_i - \bar{x}_A\|_2^2}. \tag{6}$$

This shared transductive probe compares all reducers. It is neither the autoencoder's native decoder error nor an out-of-corpus generalization test.

3.4 Labels and article groups

For arXiv, categories occurring at least five times in the fixed metric sample are kept, leaving 4,857 rows and 71 categories. A fixed stratified 80/20 split trains a distance-weighted 16-neighbor classifier in layout space. We report ordinary accuracy and balanced accuracy, the mean per-category recall. Labels are never supplied to a reducer.

For Wikipedia, a URL identifies paragraphs from the same article. The 3,000 fixed anchors query the complete one-million-row layout. Let G_i be the rows sharing anchor i's URL, and let $m_i = |N_Y^{16}(i) \cap G_i|$. Empty and full-corpus-singleton groups are excluded, leaving an eligible set E of 2,895 anchors. We report

$$\text{Purity@16} = \frac{1}{|E|} \sum_{i \in E} \frac{m_i}{16}, \tag{7}$$

$$\text{Hit@16} = \frac{1}{|E|} \sum_{i \in E} \mathbf{1}[m_i > 0], \tag{8}$$

$$\text{TrRecall@16} = \frac{1}{|E|} \sum_{i \in E} \frac{m_i}{\min(16, |G_i| - 1)}. \tag{9}$$

The last quantity is explicitly a *truncated* recall: it can reach one for a large article after retrieving 16 mates, even if additional same-article paragraphs remain.

3.5 Native autoencoder reconstruction

For the final refitted model, native reconstruction is summarized by

$$\text{NMSE}_{\text{AE}} = \frac{\sum_i \|x_i - g_\phi(f_\theta(x_i))\|_2^2}{\sum_i \|x_i - \bar{x}\|_2^2}.$$

Writing $\hat{x}_i = g_\phi(f_\theta(x_i))$, mean reconstruction cosine is

$$\text{ReconCos}_{\text{AE}} = \frac{1}{N} \sum_{i=1}^{N} \frac{\hat{x}_i^\top x_i}{\max(\|\hat{x}_i\|_2, 10^{-12})}.$$

The usual factor $\|x_i\|_2$ is one because all benchmark rows are nonzero and normalized. These are autoencoder-only diagnostics; Equation (6) is the cross-method comparison.

3.6 Controlled intrinsic-dimension cross-check

The fixed-width benchmark asks how well a chosen d preserves structure; it does not itself estimate an intrinsic dimension. We therefore run a separate controlled check on three 600-point manifolds whose local dimensions are known: a rank-three Gaussian linear subspace, a circle, and a noise-free Swiss roll. Each is orthogonally embedded in $\mathbb{R}^{10}$, independently for seeds 0, 1, and 2. Exact Euclidean input-space neighbors are used throughout. No DiRe layout is fitted in this check: the connection to DiRe is that the same neighbor-distance calculation can be applied to the input-space k-NN graph that DiRe has already constructed.

For TwoNN [4], let r_{i1} and r_{i2} be the first two neighbor radii and $\mu_i = r_{i2}/r_{i1}$. If F is the empirical CDF of the valid sorted ratios, the ideal local model obeys

$$-\log(1 - F(\mu)) = d \log \mu. \tag{10}$$

Writing $x_j = \log \mu_j$ and $y_j = -\log(1 - F_j)$, our estimate is the through-origin least-squares slope

$$\widehat{d}_{\text{TwoNN}} = \frac{\sum_{F_j < 0.9} x_j y_j}{\sum_{F_j < 0.9} x_j^2}. \tag{11}$$

Thus the noisiest upper 10% of the ratio CDF is excluded. As a second local estimate, the Levina–Bickel MLE with the MacKay–Ghahramani inverse averaging correction [5] uses $k = 20$:

$$\widehat{d}_{\text{MLE}} = \left[\frac{1}{N} \sum_{i=1}^{N} \frac{1}{k-1} \sum_{j=1}^{k-1} \log \frac{T_{ik}}{T_{ij}} \right]^{-1}. \tag{12}$$

All 600 rows in every dataset–seed cell pass the positive-distance and nondegenerate- ratio checks.

The reconstruction estimate uses a separate symmetric $10 \to 128 \to 64 \to q \to 64 \to 128 \to 10$ autoencoder for each $q \in \{1, \ldots, 6\}$. Hidden layers use SiLU and the bottleneck and output are linear. After centering by the training mean and dividing all features by one global training RMS, an 80/20 split supplies training and validation rows. AdamW uses learning rate 10^{-3}, weight decay 10^{-5}, batch size 256, a 100-epoch cap, and patience 12. Within a seed, every width uses the same split. If $\widehat{X}_{q,t}$ is the validation prediction after epoch t, the recorded normalized validation error is

$$e_q = \min_t \frac{\|X_{\text{val}} - \widehat{X}_{q,t}\|_F^2}{\|X_{\text{val}} - \bar{X}_{\text{train}}\|_F^2}. \tag{13}$$

The validation rows therefore support both early stopping and the model-selection curve; they are not an untouched test set. Each seed contributes one fit per width. The six errors are projected onto a non-increasing sequence $\widetilde{e}_q$ by isotonic regression. With

$$u_q = \frac{q-1}{5}, \qquad v_q = \frac{\widetilde{e}_q - \widetilde{e}_6}{\widetilde{e}_1 - \widetilde{e}_6},$$

the chord-elbow score is $h_q = (1 - u_q) - v_q$, and only its largest interior value is a candidate. It is accepted when $\max h_q \geq 0.05$, the relative error drop is at least 0.05, and the decrease from $q = 5$ to $q = 6$ is at most 0.01 of the total decrease. These guards prevent a flat curve or a curve still falling at the grid boundary from being reported as a dimension.

TwoNN and MLE estimate local dimension from neighbor-volume growth. The autoencoder elbow instead estimates a global, architecture- and optimization-dependent latent width for reconstruction. Their numerical comparison is therefore a cross-check between related questions, not an assertion that they are interchangeable estimators.

3.7 Sampled topology descriptions

Three label-stratified 4,000-paper arXiv samples and three uniform 3,000-paragraph Wikipedia samples are used. For each input or layout cloud Z, define its exact sampled diameter $D_Z = \max_{i,j} \|z_i - z_j\|_2$ and rescale $Z' = Z/D_Z$. Ripser persistence diagrams [7] through dimension one define Betti curves

$$\beta_q^Z(t) = \#\{(b, d) \in \mathcal{D}_q(Z') : b \leq t < d\}, \qquad q \in \{0, 1\}. \tag{14}$$

Each cloud is sampled at 50 filtration values extending to 1.05 times that cloud's own longest finite death. The reported quantity is the unnormalized FastDTW distance [8], using default radius one and local cost $|u - v|$, between the two 50-value β_q sequences. The filtration axes themselves are not supplied to FastDTW and generally differ between input and layout.

Consequently, Betti-curve DTW is a same-corpus, sampled descriptive discrepancy—not a proof of homology preservation. The nine values per method cross three reducer seeds with three evaluation samples and are not nine independent replicates. For finite positive H_0 persistences $p_1, \ldots, p_m$, supplementary "islandness" descriptions are the longest bar $\max_j p_j$, the mean of the five largest bars divided by the median, and the Gini coefficient

$$G(p) = \frac{\sum_{i=1}^m \sum_{j=1}^m |p_i - p_j|}{2m \sum_{j=1}^m p_j}.$$

These are descriptions, not distances to the input, and have no universal better direction.

4 Results

Boldface in the following tables marks only the best observed three-seed mean; it is not a significance claim. With three seeds, the main rankings are generally stable but uncertainty estimates are necessarily descriptive.

4.1 **arXiv:** $723{,}457 \times 384 \to \{2, 8\}$

d	Method	Fit (s)↓	Trust↑	Cont.↑	Overlap↑	ρ_S ↑	Stress↓	Decode↓	Accuracy↑	Bal. acc.↑
2	Autoencoder	24.32	**.877**	**.918**	.099	**.588**	.523	.647	.363	.139
2	DiRe	**15.33**	.792	.860	.116	.313	.426	.721	.377	.146
2	UMAP	30.03	.858	.902	**.147**	.489	**.418**	**.643**	**.482**	**.245**
8	Autoencoder	23.42	**.969**	**.988**	**.321**	**.801**	**.254**	**.466**	.509	.250
8	DiRe	**15.08**	.905	.897	.187	.355	.335	.534	.531	.254
8	UMAP	30.66	.937	.954	.247	.590	.357	.499	**.609**	**.329**

Table 2: arXiv means over three reducer/training seeds at both target widths. Boldface is assigned within each width. "Decode" is the shared local-decoding NMSE, not native autoencoder reconstruction.

At width two, the autoencoder is strongest on rank-aware local fidelity and pair-distance ordering, UMAP retains the most exact neighbors and predicts primary category best, and UMAP and DiRe give the two lowest stresses. At width eight, the autoencoder leads all six geometric and decoding columns, while UMAP remains clearly best on ordinary and balanced category accuracy. Increasing width from two to eight raises the autoencoder's neighbor overlap from .099 to .321 and lowers its decoding NMSE from .647 to .466; the corresponding category accuracy rises from .363 to .509 but remains below UMAP's .609. This is a within-corpus width contrast, although two tested widths do not establish a general scaling law. Figure 1 shows the seed-0 native width-two layouts and secondary PCA views of width eight, but the classifier scores in Table 2 are the quantitative label test.

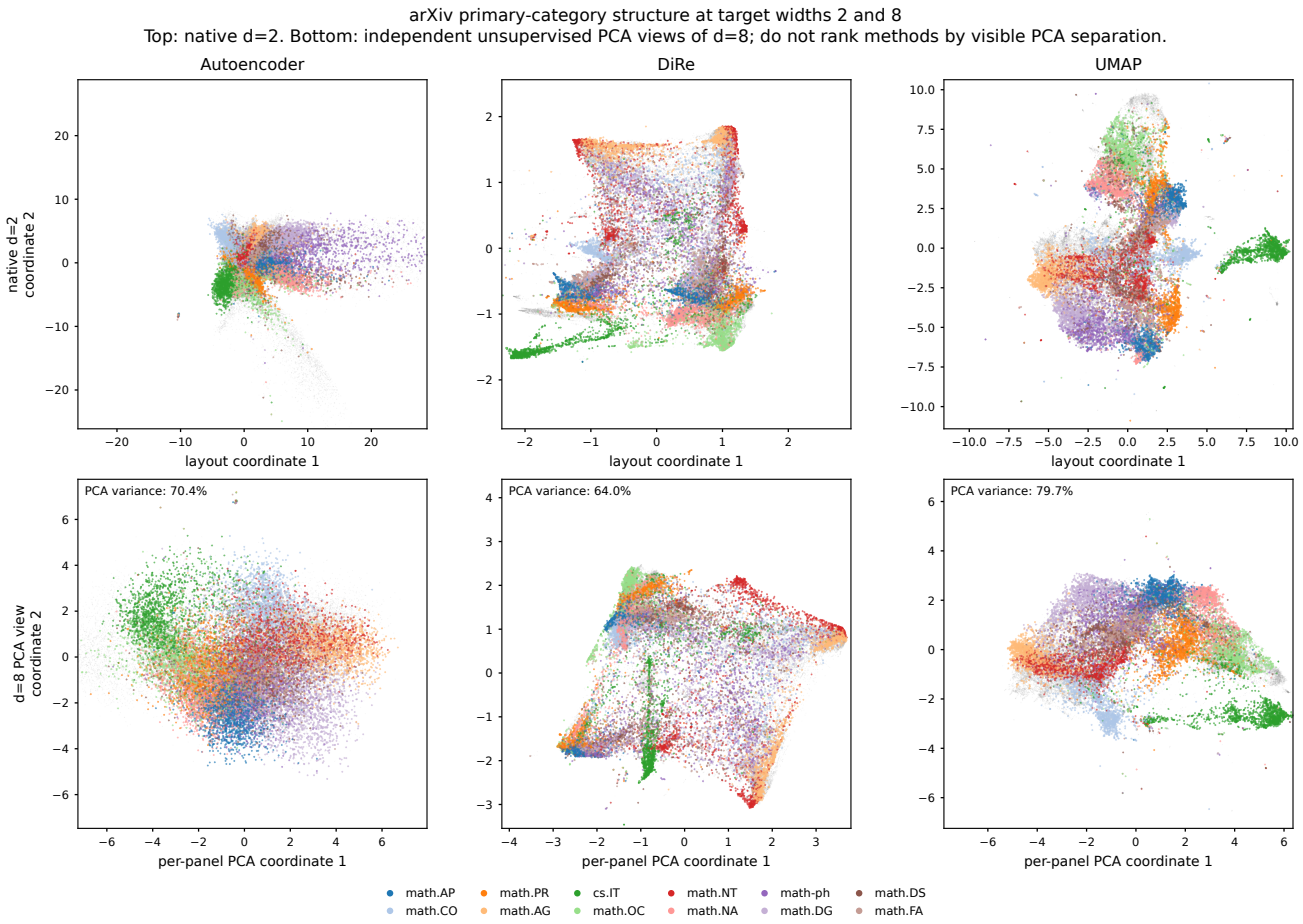


Figure 1: Seed-0 arXiv layouts at both target widths. The top row is native $d = 2$; the bottom row is an independent unsupervised PCA view of each native $d = 8$ layout, fitted on the same uniform 100,000 rows. Every panel uses 1,500 papers from each of the 12 largest primary categories and a uniform 30,000-paper background outside those categories. Colors and displayed indices are identical across widths and methods. Coordinates and isotropic robust limits are panel-specific; visible separation in the PCA row is not a cross-method score.

4.2 Wikipedia: $10^6 \times 1024 \to \{2, 8\}$

d	Method	Fit (s)↓	Trust↑	Cont.↑	Overlap↑	ρ_S ↑	Stress↓	Decode↓	Purity↑	Hit↑	TrRecall↑
2	Autoencoder	81.60	**.887**	.827	**.183**	.230	.747	.907	.006	.050	.006
2	DiRe	**62.14**	.752	.823	.118	.305	**.428**	.959	.136	.288	.141
2	UMAP	144.01	.847	**.832**	.182	**.337**	.475	**.907**	**.201**	**.498**	**.236**
8	Autoencoder	81.85	**.960**	**.944**	**.337**	**.459**	.362	**.791**	.071	.283	.079
8	DiRe	**63.93**	.908	.894	.229	.378	**.268**	.844	.306	.570	.345
8	UMAP	144.05	.917	.885	.235	.389	.438	.841	**.410**	**.754**	**.530**

Table 3: Wikipedia means over three reducer/training seeds at both target widths. Boldface is assigned within each width. Article purity, hit rate, and truncated recall query 3,000 anchors against the complete one-million-row layout. At width two, UMAP's mean decoding NMSE is only 0.00003 below the autoencoder's before rounding.

At width two, the autoencoder gives the highest trustworthiness and a narrowly higher exact-neighbor overlap than UMAP; UMAP leads continuity, pair-distance ordering, the shared decoding probe, and all three article-group scores; and DiRe gives the lowest stress. At width eight, the autoencoder wins every sample-conditional local-geometry metric and the shared decoding probe. It does not organize exact article groups as strongly: UMAP places at least one same-article paragraph among the first 16 neighbors for 75.4% of eligible anchors, compared with 57.0% for DiRe and 28.3% for the autoencoder. DiRe gives the lowest global stress at both widths. Wikipedia thus has meaningful group labels, but they express exact article identity rather than broad semantic categories. The winning method for every Table 3 width-eight metric is unchanged across the three seeds; the largest listed sample standard deviation is 0.018 for autoencoder stress.

All three reducers improve every reported Wikipedia quality metric when moving from two to eight coordinates. The autoencoder changes most sharply: stress falls from .747 to .362, pair correlation doubles from .230 to .459, and article hit rate rises from .050 to .283. UMAP remains the article-group leader at both widths, reaching .754 hit rate at width eight. The crossover therefore separates a target-width effect from the earlier corpus comparison: the autoencoder's width-eight local-geometry advantage is not merely a peculiarity of using Wikipedia rather than arXiv.

Figure 2 shows the native width-two layouts and illustrative PCA views of width eight. The PCA panels' axes and visible separations cannot be treated as the scored embedding. Figure 3 shows the native-layout retrieval statistics over both widths and all three reducer seeds.

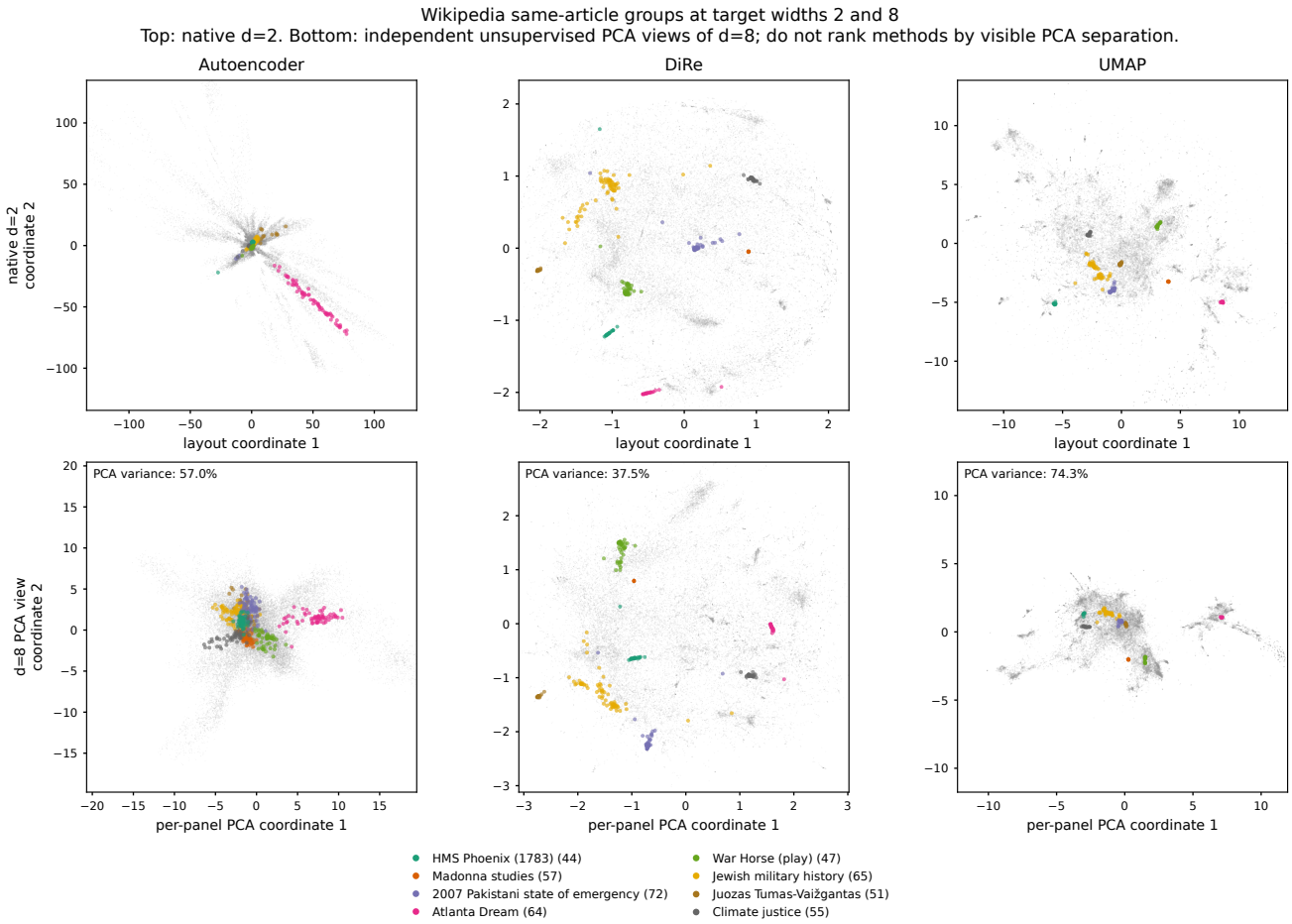


Figure 2: Seed-0 Wikipedia layouts at both target widths. The top row is native $d = 2$; the bottom row is an independent unsupervised PCA view of each native $d = 8$ layout, fitted on the same uniform 100,000 rows. Eight articles containing 40–100 paragraphs are chosen by a deterministic prespecified rule independent of all layouts, and every paragraph from them is shown over the same 30,000-row background. Highlighted and background indices are identical across widths and methods. Coordinates and isotropic robust limits are panel-specific; displayed PCA variance is descriptive, and no cross-method ranking should be inferred from the PCA row.

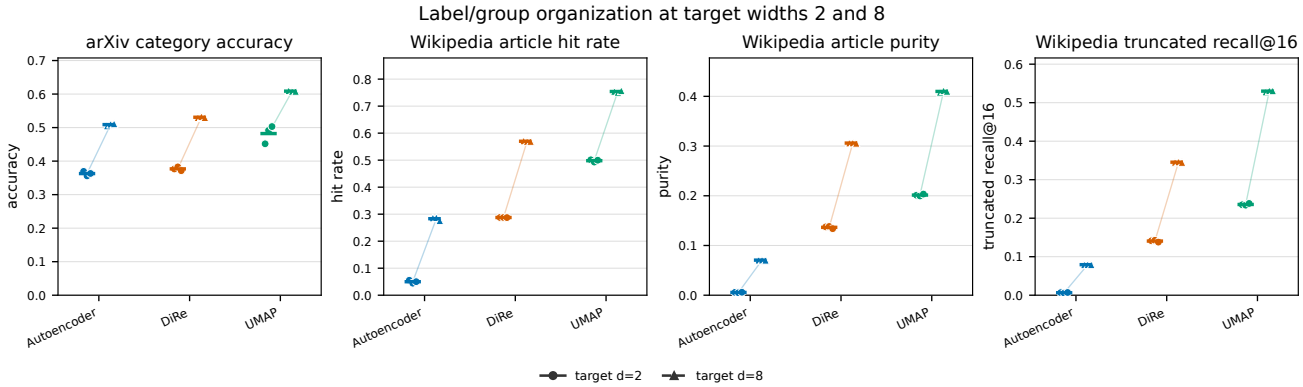


Figure 3: Label/group organization in the native layouts. Points are the three reducer/training seeds and horizontal segments are means; circles and solid mean bars are $d = 2$, triangles and dashed bars are $d = 8$, and faint segments join width means. arXiv uses category classification accuracy. Wikipedia uses full-million-layout article hit rate, purity, and truncated recall at 16 with the same 3,000 anchors at both widths. These are native-layout metrics, not scores of the PCA views in Figure 2. Labels are evaluation-only and are not supplied to any reducer.

4.3 Sampled topology descriptions

Corpus	d	Method	FastDTW(β_0)↓	FastDTW(β_1)↓
arXiv	2	Autoencoder	4,841	439
	2	DiRe	**3,322**	**176**
	2	UMAP	10,414	941
	8	Autoencoder	**1,583**	3,045
	8	DiRe	2,685	**177**
	8	UMAP	6,469	499
Wikipedia	2	Autoencoder	4,822	**8,974**
	2	DiRe	**2,107**	9,355
	2	UMAP	3,566	9,207
	8	Autoencoder	**1,591**	**7,572**
	8	DiRe	3,335	9,117
	8	UMAP	2,986	9,198

Table 4: Pooled descriptive means across three reducer seeds and three evaluation samples. Lower raw sequence discrepancy is preferable within a corpus and target width, subject to the sampling, cloud-specific filtration grids, and FastDTW qualifications in the text.

At arXiv width two, DiRe has the lowest sampled β_0 and β_1 curve discrepancies, and the autoencoder lies between DiRe and UMAP. At width eight the result splits: the autoencoder has the lowest β_0 discrepancy, while DiRe remains lowest on β_1. Increasing arXiv width lowers β_0 discrepancy for all three methods, but β_1 is nearly unchanged for DiRe, improves for UMAP, and worsens substantially for the autoencoder. Wikipedia behaves differently. At width two, DiRe has the lowest β_0 discrepancy and the autoencoder the lowest β_1 discrepancy; at width eight the autoencoder has the lowest value in both columns. From width two to eight, both autoencoder discrepancies fall, whereas DiRe's β_0 discrepancy rises and UMAP's β_1 discrepancy is essentially unchanged. Thus neither a wider target nor one reducer has a corpus-independent topological advantage in these samples; the rankings exhibit a corpus-by-width interaction. Table 4 is a descriptive sample comparison, not a claim that the reported values prove preservation of the full-corpus homology.

4.4 Local and reconstructive dimension estimates are related, not identical

Manifold	Known local d	TwoNN	MLE ($k = 20$)	AE elbow
Rank-three linear	3	3.036 ± 0.094	3.053 ± 0.045	3.000 ± 0.000
Circle	1	1.034 ± 0.061	0.997 ± 0.008	2.000 ± 0.000
Swiss roll	2	1.990 ± 0.084	1.929 ± 0.033	2.667 ± 0.577

Table 5: Controlled input-space intrinsic-dimension cross-check. Entries are mean±sample standard deviation across three independent data/model seeds. The AE elbows for the Swiss roll are 3, 3, and 2. These synthetic results are not estimates for the arXiv or Wikipedia corpora.

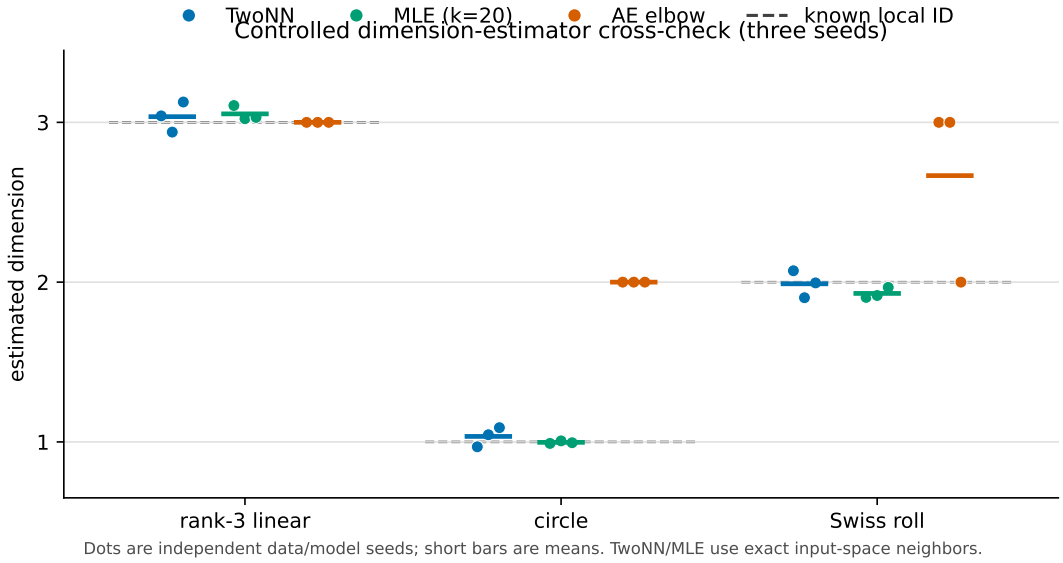


Figure 4: The three seeded estimates underlying Table 5. Dots show individual seeds, short horizontal segments show means, and dashed segments show known local dimension. TwoNN and MLE use exact input-space neighbors; the AE points are accepted validation-reconstruction elbows.

Rounding TwoNN to the nearest integer, its estimate and the AE elbow agree exactly in four of nine seeded trials and differ by at most one in all nine. Their paired Spearman correlation is $\rho_S = .779$. This correlation is descriptive only: the nine pairs are clustered within three deliberately chosen manifold families, so neither a null-model p-value nor a population-level inference is warranted.

The pattern is more informative than the aggregate count. Both estimators recover the rank-three linear subspace. The circle is locally one-dimensional, but a continuous Euclidean autoencoder generally needs two coordinates to reconstruct it globally without a cut, so its stable 1 versus 2 difference reflects the two estimands rather than a failure. A Swiss roll is locally two-dimensional and globally unrollable into a plane; TwoNN recovers two, while finite-network optimization together with the chord-elbow rule selects either two or three across seeds. Thus the pointwise and neighbor-based approaches do not produce unrelated notions of dimension, but topology, model capacity, optimization, and elbow-rule sensitivity prevent an identity theorem.

4.5 Why is the autoencoder so fast?

Corpus	d	GPU staging	Epoch selection	Full refit	Encode/statistics	Total
arXiv	2	0.14	11.90	11.58	0.69	24.32
arXiv	8	0.11	11.43	11.21	0.67	23.42
Wikipedia	2	0.40	39.14	39.60	2.46	81.60
Wikipedia	8	0.42	39.29	39.67	2.47	81.85

Table 6: Mean autoencoder fit-time decomposition in seconds. The headline time includes both training phases.

The timing is less paradoxical than it first appears. The arXiv and Wikipedia networks are small, their complete 1.1 and 4.1 GB float32 matrices fit in GPU memory, and a batch of 32,768 leaves only about 23 and 31 steps per epoch. Approximately 400 epochs across selection and refitting therefore require about 8,800 and 12,000 regular dense optimizer steps after accounting for the 5% selection holdout. The primary results in Table 6 use no mixed precision and include both the validation-selection model and the fresh full-data model.

These are implementation measurements on a GH200, not hardware-independent complexity constants. The graph reducers also avoid quadratic behavior, but do so through approximate indexing and sparse operations rather than dense GEMMs.

4.6 BF16 sensitivity: tested margins pass, topology is mixed, and speed is model-size dependent

We repeated every autoencoder cell—both corpora, $d \in \{2, 8\}$, and seeds 0–2—with CUDA BF16 autocast during training, epoch-selection validation, and final encoding. Parameters, gradients, AdamW state, the explicit MSE loss, and reconstruction accumulations remained float32. Inputs, samples, seeds, architecture, and optimizer settings matched the primary runs. These are paired sensitivity runs; the scientific comparison tables remain the primary float32 results.

		Selection + refit training			Complete reported fit			Native NMSE
Corpus	d	FP32 (s)	BF16 (s)	Speedup	FP32 (s)	BF16 (s)	Speedup	BF16/FP32
arXiv	2	23.49	25.49	0.920×	24.32	26.28	0.925×	1.00193
arXiv	8	22.64	25.46	0.889×	23.42	26.25	0.892×	1.00131
Wikipedia	2	78.74	39.43	1.997×	81.60	42.28	1.931×	1.00096
Wikipedia	8	78.96	41.53	1.901×	81.85	44.34	1.846×	1.00023

Table 7: Paired three-seed BF16 sensitivity. Times are arithmetic means. Speedup is FP32 time divided by BF16 time; training speedup is the paired geometric mean and total speedup is the paired arithmetic mean. The displayed NMSE ratio is the arithmetic mean of the three paired ratios. Values below one indicate a BF16 slowdown.

Both the standard and FP32-evaluation-isolation BF16 arms pass all prespecified practical checks on native reconstruction and shared geometry or label metrics in every cell; with only three seeds these margins are descriptive, not formal equivalence tests. The largest paired BF16/FP32 native-NMSE ratio in the standard arm is 1.00243. The sampled-topology response is mixed rather than uniformly invariant. Pooled (β_0, β_1) DTW means change from $(4841, 439)$ to $(5660, 537)$ for arXiv $d = 2$, from $(1583, 3045)$ to $(1659, 3068)$ for arXiv $d = 8$, from $(4822, 8974)$ to $(4820, 8986)$ for Wikipedia $d = 2$, and from $(1591, 7572)$ to $(1505, 8121)$ for Wikipedia $d = 8$. A clean rerun that trained with BF16 but performed validation, encoding, reconstruction, shared-metric evaluation, and topology input generation in float32 left shared-metric quality essentially unchanged and moved the pooled topology mean closer to FP32 in only four of eight corpus–width–Betti comparisons. Thus this descriptive isolation does not support final-coordinate quantization as the main explanation for the topology differences.

A 100-step resident-batch microprofile explains why a universal three- or four-fold speedup does not appear. With TF32 matrix multiplication disabled, this fixed-order profile uses a synthetic standard-normal resident batch and separate warm-ups for each precision. Its FP32/BF16 total-step speedups are 0.859 and 0.842 for the two arXiv widths, versus 2.407 and 2.428 for Wikipedia; the Wikipedia forward-plus-backward portion alone reaches about 2.65×. The approximately 272,000-parameter arXiv MLP is overhead-bound at this batch shape, so autocast dispatch and conversions outweigh faster eligible kernels. The roughly 1.35-million-parameter Wikipedia MLP supplies enough dense matrix work to benefit, although FP32 AdamW and the non-training parts of the fit cap the end-to-end gain near 1.85–1.93×. Because the full-fit FP32 runs and later BF16 runs were not alternated contemporaneously, their timing ratios are descriptive; the microprofile independently reproduces the workload crossover.

4.7 Longer MSE optimization is not a geometry rule

For these two original cells, reconstruction convergence is therefore not a universal stopping criterion for a useful layout. Longer training improves native MSE, trustworthiness, and exact overlap, but reduces continuity and pair-rank correlation, worsens stress, and gives mixed or worse sampled Betti-curve discrepancies. No analogous

Corpus	Run	Native NMSE↓	Trust↑	Cont.↑	Overlap↑	ρ_S ↑	Stress↓	DTW β_0 ↓	DTW β_1 ↓
arXiv	primary (199 epochs)	.5965	.8818	.9140	.1006	.5925	.5615	4,544	516
arXiv	longer (398 epochs)	.5753	.8940	.9091	.1135	.5746	.6955	5,771	772
Wikipedia	primary (200 epochs)	.6910	.9600	.9426	.3339	.4593	.3818	1,599	8,146
Wikipedia	longer (400 epochs)	.6744	.9629	.9396	.3460	.4517	.3969	1,545	9,709

Table 8: Reducer-seed-0 400-epoch-cap sensitivity checks for the two original benchmark cells only: arXiv $d = 2$ and Wikipedia $d = 8$. DTW columns average the three fixed evaluation samples. Longer training lowers native reconstruction error and improves some local scores, while several global and sampled-topology scores worsen.

400-epoch claim is made for the crossed target widths. A geometry-aware autoencoder objective might change this tradeoff, but it would also move away from the deliberately simple pointwise MSE experiment studied here.

5 Conclusions and limitations

The reducer results support six narrow conclusions.

1. A small MSE autoencoder is computationally practical at 723,000–1,000,000 rows in this GH200 configuration. Its favorable timing is consistent with fixed-architecture linear-in-N pointwise work and efficient dense kernels. BF16 autocast passes the tested reconstruction and shared-metric practical margins, while its sampled-topology response is mixed. Its speed is shape-dependent: it nearly halves Wikipedia training time and slightly slows the smaller arXiv model.
2. Plain reconstruction can preserve sampled local geometry and decodability very well without an explicit pairwise objective. This does not guarantee strong label/group organization: UMAP wins arXiv category prediction at both widths and Wikipedia same-article retrieval at both widths.
3. On both corpora, increasing d from two to eight changes local-fidelity rankings as well as score magnitudes: the autoencoder leads the local, distance-order, and decoding columns at width eight but not at width two. DiRe retains the best Wikipedia stress. This repeated contrast supports a target-width effect within the tested settings, not a general scaling law.
4. Controlled known-manifold checks show approximate rather than exact agreement between TwoNN local dimension and the AE reconstruction elbow. Their systematic circle difference illustrates why local and global reconstructive dimension must be interpreted separately.
5. More MSE optimization is not automatically better dimension reduction. Reconstruction, local ranks, global distances, labels, and sampled topology can move in different directions in the two original sensitivity cells.
6. Sampled topology has no universal method or width ordering here. The arXiv width-eight autoencoder improves markedly in β_0 but worsens in β_1; the Wikipedia width-eight autoencoder improves in both. These completed cells point to a corpus-by-width interaction rather than a single topological winner.

The comparison has important limits. It uses one accelerator, one architecture family, one neighborhood size, and only two target widths. arXiv category labels and Wikipedia URL groups have different meanings. All primary layouts are transductive, and the autoencoder is retrained for every corpus, width, and seed. Neighborhood metrics are sample-conditional; distance metrics use sampled pairs; topology uses small sampled clouds and cloud-specific curve grids. The three-seed comparisons are descriptive, not formal uncertainty quantification, and two target widths cannot identify a general dimension-response curve. The controlled dimension check uses small synthetic manifolds and does not estimate the intrinsic dimension of either text corpus.

Reproducibility

The companion *dimension-reduction-benchmarks* repository [11] provides the benchmark, topology, and controlled-dimension implementations together with compact result tables and machine-readable figure provenance. The supplementary records retain exact environment information, checksums, sample identities, and complete topology curves; those audit identifiers are intentionally kept out of the public-facing narrative.

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